

Title: Database Systems Exam

Course Code: IT301

Instructor: Dr. Asma

Duration: 30

Q1. What is a primary key in a relational database? (mcq) [1]

- a) A key that can have duplicate values
- b) A unique identifier for each record in a table
- c) A key used only for indexing
- d) A foreign key reference

Correct Answer: B

Q2. Which SQL command is used to retrieve data from a database? (mcq) [1]

- a) INSERT
- b) UPDATE
- c) SELECT
- d) DELETE

Correct Answer: C

Q3. What does SQL stand for? (mcq) [1]

- a) Structured Query Language
- b) Simple Question Language
- c) Standard Query Logic
- d) System Query Layer

Correct Answer: A

Q4. Which of the following is a NoSQL database? (mcq) [1]

- a) MySQL
- b) PostgreSQL
- c) MongoDB
- d) Oracle

Correct Answer: C

Q5. What is a foreign key? (mcq) [1]

- a) A key that encrypts data in a table
- b) A field that links two tables by referencing the primary key of another table
- c) A key used for sorting records
- d) A key that allows duplicate entries only

Correct Answer: B

Q6. Which normal form eliminates partial dependencies? (mcq) [1]

- a) First Normal Form
- b) Second Normal Form
- c) Third Normal Form
- d) Boyce-Codd Normal Form

Correct Answer: B

Q7. What is the purpose of an index in a database? (mcq) [1]

- a) To delete duplicate records automatically
- b) To speed up data retrieval operations
- c) To encrypt sensitive data
- d) To define relationships between tables

Correct Answer: B

Q8. Which SQL clause is used to filter records after grouping? (mcq) [1]

- a) WHERE
- b) ORDER BY
- c) HAVING
- d) GROUP BY

Correct Answer: C

Q9. A database transaction must follow the ACID properties. (true_false) [0.5]

- a) True
- b) False

Correct Answer: A

Q10. The WHERE clause in SQL is used to filter records before grouping. (true_false) [0.5]

- a) True
- b) False

Correct Answer: A

Q11. A table can have more than one primary key. (true_false) [0.5]

- a) True
- b) False

Correct Answer: B

Q12. DDL stands for Data Definition Language. (true_false) [0.5]

- a) True
- b) False

Correct Answer: A